

Estimate Key COGS Components (Average \$40,000 Car Estimate)

A. Battery Costs

- Industry benchmark: Battery costs in 2024 were around **\$130/kWh**, with GM targeting **\$100/kWh** for its Ultium platform.
- Example: For an average vehicle with a 65 kWh battery pack, the battery cost would be:
- $65\text{kWh} \times \$130/\text{kWh} \sim \mathbf{\$8,500}$

B. Chassis, Body, and Powertrain

- Chassis and body costs are typically **25-30% of total COGS**.
- Electric powertrain (motor, inverter) costs are **10-20% of total COGS**.
- For a mid-sized EV priced at \$40,000, chassis and powertrain could total around **\$8,000–\$12,000**.

C. Labor Costs

- Labor costs in the U.S. are estimated at **\$1,500–\$2,500 per vehicle**, depending on automation levels.

D. Other Components

- Additional costs (software, HVAC, infotainment, tires, etc.) are around **10-15% of COGS**.
- **Estimate: \$2,000–\$4,000 per vehicle.**

Total COGS: ~\$25,500 with current state of economy; 63.75% selling price

In the fourth quarter of 2024, GM announced that its EV lineup became "variable profit positive." This means that the revenue from EV sales exceeded the variable costs associated with producing them, such as labor and materials

GM has been actively working to reduce battery costs, a significant component of EV COGS. The company aims to lower battery costs to approximately \$100 per kilowatt-hour with its Ultium battery technology.